

Calculus I Lesson 33

Definition of Antiderivative:

A function $F(x)$ is an antiderivative of a function $f(x)$ if $F'(x) = f(x)$.

1) Let $f(x) = x + 7$

Notation for antiderivative: $F(x) = \int f(x) \cdot dx$

a) Is $F(x) = \frac{x^2}{2} + 7x + 2$ is an antiderivative of $f(x) = x + 7$?

b) Is $F(x) = \frac{x^2}{2} + 7x + 5$ is an antiderivative of $f(x) = x + 7$?

c) Is $F(x) = \frac{x^2}{2} + 7x - 7$ is an antiderivative of $f(x) = x + 7$?

d) Is $F(x) = \frac{x^2}{2} + 7x + C$ is an antiderivative of $f(x) = x + 7$? C is a constant.

2) Let $f(x) = x^5 + 5$

Notation for antiderivative: $F(x) = \int f(x) \cdot dx$

a) Is $F(x) = \frac{x^6}{6} + 5x + 1$ is an antiderivative of $f(x) = x^5 + 5$?

b) Is $F(x) = \frac{x^6}{6} + 5x + 3$ is an antiderivative of $f(x) = x^5 + 5$?

c) Is $F(x) = \frac{x^6}{6} + 5x - 8$ is an antiderivative of $f(x) = x^5 + 5$?

d) Is $F(x) = \frac{x^6}{6} + 5x + C$ is an antiderivative of $f(x) = x^5 + 5$? C is a constant.

3) Let $f(x) = x^2 + 4x + 5$

Notation for antiderivative: $F(x) = \int f(x) \cdot dx$

a) Is $F(x) = \frac{x^3}{3} + 2x^2 + 5x - 9$ is an antiderivative of $f(x) = x^2 + 4x + 5$?

b) Is $F(x) = \frac{x^3}{3} + 2x^2 + 5x - 1$ is an antiderivative of $f(x) = x^2 + 4x + 5$?

c) Is $F(x) = \frac{x^3}{3} + 2x^2 + 5x + 5$ is an antiderivative of $f(x) = x^2 + 4x + 5$?

d) Is $F(x) = \frac{x^3}{3} + 2x^2 + 5x + C$ is an antiderivative of $f(x) = x^2 + 4x + 5$? C is a constant.

4) Let $f(x) = x^{3/2} + 2x - 9$

Notation for antiderivative: $F(x) = \int f(x) \cdot dx$

a) Find an antiderivative of $f(x) = x^{3/2} + 2x - 9$.

$$F(x) = \int f(x) \cdot dx = \underline{\hspace{2cm}}?$$

b) Is $F'(x) = f(x)$?

5) Let $f(x) = \sqrt[3]{x^4} = x^{4/3}$

Notation for antiderivative: $F(x) = \int f(x) \cdot dx$

a) Find an antiderivative of $f(x) = \sqrt[3]{x^4}$.

$$F(x) = \int f(x) \cdot dx = \underline{\hspace{2cm}}?$$

b) Is $F'(x) = f(x)$?

6) Let $f(x) = \frac{1}{x^4} = x^{-4}$

Notation for antiderivative: $F(x) = \int f(x) \cdot dx$

a) Find an antiderivative of $f(x) = \frac{1}{x^4}$.

$$F(x) = \int f(x) \cdot dx = \underline{\hspace{2cm}} ?$$

b) Is $F'(x) = f(x)$?

7) Let $f(x) = \frac{x+12}{\sqrt{x}}$

Hint: $\frac{x+12}{\sqrt{x}} = \frac{x+12}{x^{1/2}} = \frac{x^1}{x^{1/2}} + \frac{12}{x^{1/2}} = x^{1/2} + 12x^{-1/2}$

Notation for antiderivative: $F(x) = \int f(x) \cdot dx$

a) Find an antiderivative of $f(x) = \frac{x+12}{\sqrt{x}}$.

$$F(x) = \int f(x) \cdot dx = \underline{\hspace{2cm}} ?$$

b) Is $F'(x) = f(x)$?

8) Let $f(x) = \frac{x^3 - 5x^2 + 6}{x^3}$

Hint: $\frac{x^3 - 5x^2 + 6}{x^3} = \frac{x^3}{x^3} + \frac{-5x^2}{x^3} + \frac{6}{x^3} = 1 - 5x^{-1} + 6x^{-3}$

a) Find an antiderivative of $f(x) = \frac{x^3 - 5x^2 + 6}{x^3}$.

$$F(x) = \int f(x) \cdot dx = F(x) = \int \left(\frac{x^3 - 5x^2 + 6}{x^3} \right) \cdot dx = \underline{\hspace{2cm}} ?$$

b) Is $F'(x) = f(x)$?

9) Let $f(x) = \sqrt{x} + \frac{3}{4\sqrt{x}}$

Hint: $\sqrt{x} + \frac{3}{4\sqrt{x}} = x^{1/2} + \frac{3}{4x^{1/2}} = x^{1/2} + \frac{3}{4} \cdot x^{-1/2}$

a) Find an antiderivative of $f(x) = \sqrt{x} + \frac{3}{4\sqrt{x}}$.

$$F(x) = \int f(x) \cdot dx = F(x) = \int \left(\sqrt{x} + \frac{3}{4\sqrt{x}} \right) \cdot dx = \underline{\hspace{2cm}} ?$$

b) Is $F'(x) = f(x)$?

10) Let $f(x) = \frac{4x^4 - 8x^3 + 6x}{\sqrt{x}}$

Hint: $\frac{4x^4 - 8x^3 + 6x}{\sqrt{x}} = \frac{4x^4}{x^{1/2}} + \frac{-8x^3}{x^{1/2}} + \frac{6x}{x^{1/2}} = 4x^{7/2} - 8x^{5/2} + 6x^{1/2}$

a) Find an antiderivative of $f(x) = \frac{4x^4 - 8x^3 + 6x}{\sqrt{x}}$.

$$F(x) = \int f(x) \cdot dx = F(x) = \int \left(\frac{4x^4 - 8x^3 + 6x}{\sqrt{x}} \right) \cdot dx = \underline{\hspace{2cm}} ?$$

b) Is $F'(x) = f(x)$?

